Department of CSE(AIML), PES University
UE26CS151A -Python for Computational Problem Solving
Laboratory Week 5

Note: Students are required to write the programs along with atleast two outputs for each experiment in the laboratory observation record. The observation record must be signed by the faculty in charge. For every laboratory session, the observation component carries **3 marks**.

Problem Statements

Q1. Sum of Elements in a Tuple (Do not use sum() function.)

Write a program to find the sum of all elements in a tuple.

Q2. Count Elements in a Tuple.

Write a program to count the number of elements in a tuple.

Q3. Add an Element to a Set

Write a program to add an element to a set.

Q4. Student Data Management Using Tuple and Set

Write a Python program to:

1. Create a tuple containing the names of five students.
2. Create a set containing the marks obtained by the students.
3. Display the tuple and set.
4. Display the number of students using len().
5. Add a new mark to the set using add().
6. Display the updated set.

Q5. Search for an Element in a Tuple

Write a program to create a tuple of integers. Accept an element from the user and check whether it is present in the tuple. If found, display its position; otherwise, display an appropriate message.

Q6. Analyze Even and Odd Numbers in a Tuple

Write a program to create a tuple of integers and count the number of even and odd elements present in the tuple.

Q7. Palindrome Check Using a Tuple

Write a program to create a tuple of characters and check whether it forms a palindrome.

Q8. Tuple Rotation

Write a program to create a tuple of integers and rotate its elements one position to the left.

Solutions

1. Sum of Elements in a Tuple.

```
t = (10, 20, 30, 40)

total = 0

for i in t:
    total += i

print("Sum =", total)
```

Output
Sum = 100

2. Count Elements in a Tuple.

```
t = (5, 10, 15, 20, 25)

count = 0

for i in t:
    count += 1

print("Number of elements =", count)
```

Output
Number of elements = 5

3. Add an Element to a Set

```
s = {10, 20, 30}
```

```
s.add(40)

print(s)
```

Output

```
{40, 10, 20, 30}
```

4. Student Data Management Using Tuple and Set

```
students = ("Anil", "Bhavya", "Charan", "Divya", "Esha")

marks = {85, 92, 78, 88, 95}

print("Student Names:", students)
print("Marks:", marks)

print("Number of Students:", len(students))

marks.add(90)

print("Updated Marks:", marks)
```

Output

```
Student Names: ('Anil', 'Bhavya', 'Charan', 'Divya', 'Esha')
Marks: {85, 92, 78, 88, 95}
Number of Students: 5
Updated Marks: {85, 90, 92, 78, 88, 95}
```

5. Search for an Element in a Tuple

```
t = (10, 20, 30, 40, 50)

num = int(input("Enter the element to search: "))

if num in t:
    print("Element found at index:", t.index(num))
else:
    print("Element not found.")
```

Output
Element found at index: 2
Enter the element to search: 30

6. Analyze Even and Odd Numbers in a Tuple

```
t = (12, 25, 18, 31, 40, 17, 22)

even = 0
odd = 0

for num in t:
    if num % 2 == 0:
        even += 1
    else:
        odd += 1

print("Number of even elements:", even)
print("Number of odd elements:", odd)
```

Output
Number of odd elements: 3
Number of even elements: 4

7. Palindrome Check Using a Tuple

```
t = ('m', 'a', 'd', 'a', 'm')

if t == t[::-1]:
    print("The tuple is a palindrome.")
else:
    print("The tuple is not a palindrome.")
```

Output
The tuple is a palindrome.

8. Tuple Rotation

```
t = (10, 20, 30, 40, 50)
```

```
rotated = t[1:] + (t[0],)

print("Original Tuple:", t)
print("Tuple after Left Rotation:", rotated)
```

Output

```
Tuple after Left Rotation: (20, 30, 40, 50, 10)
Original Tuple: (10, 20, 30, 40, 50)
```

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